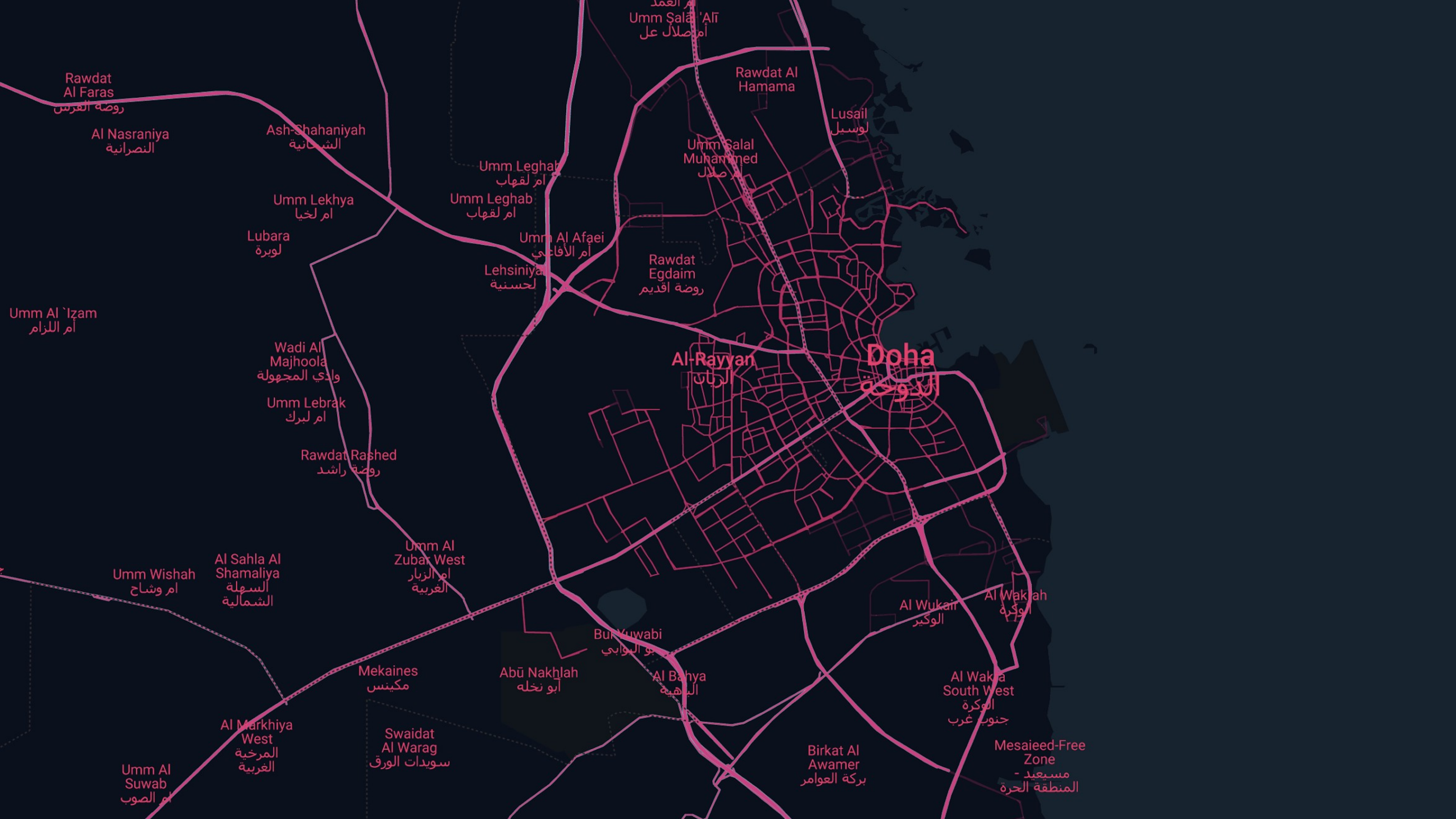
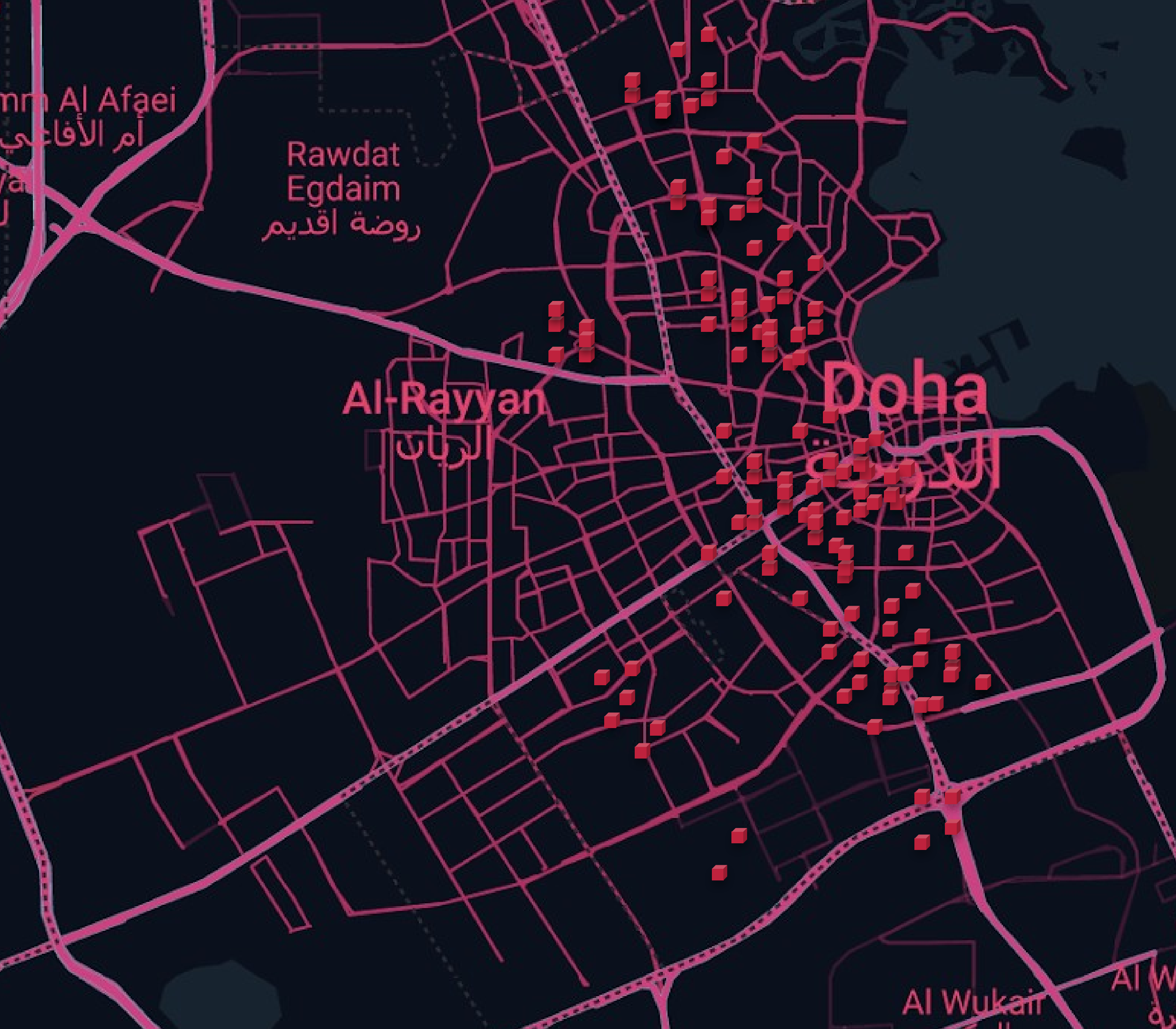
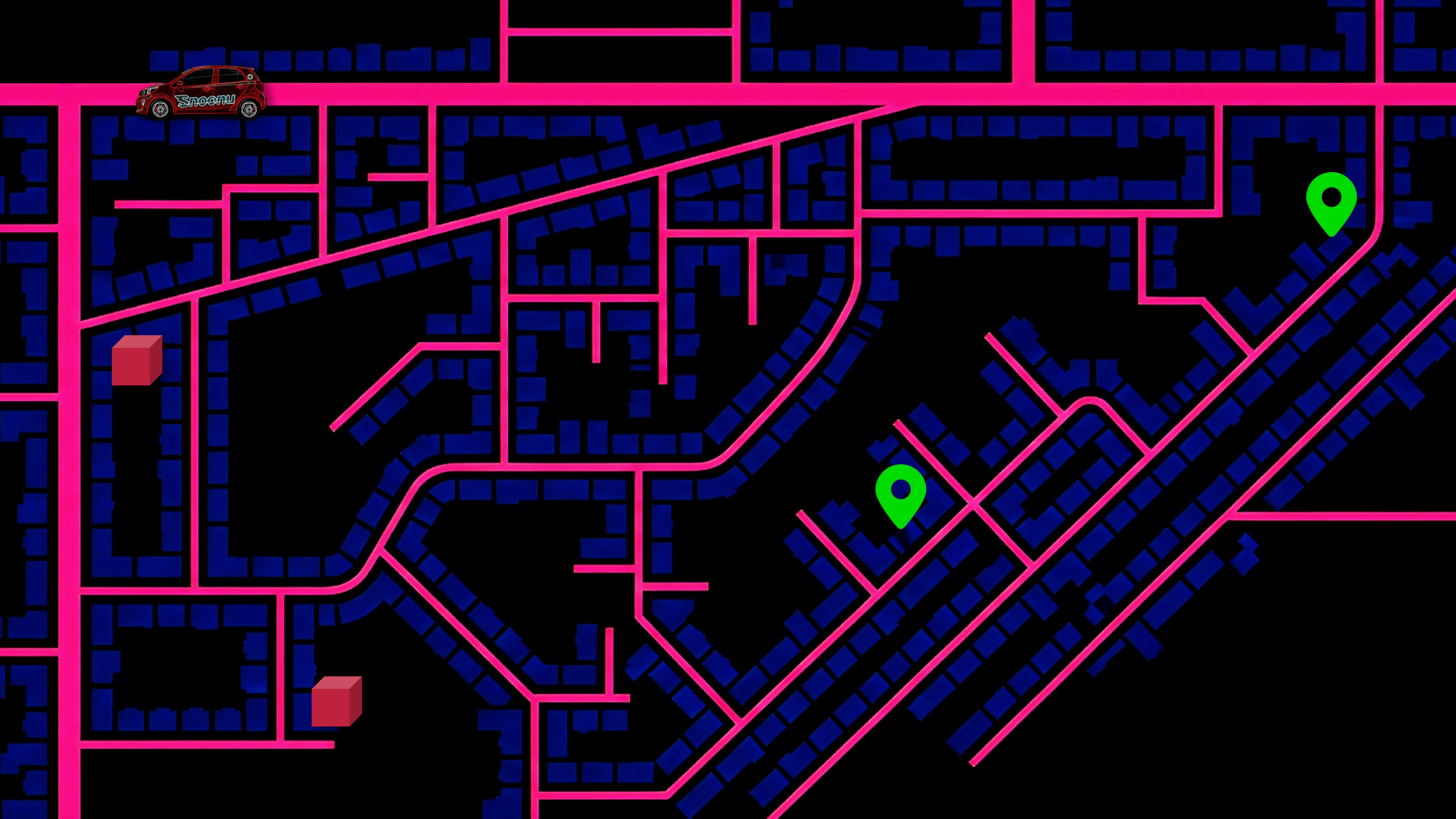
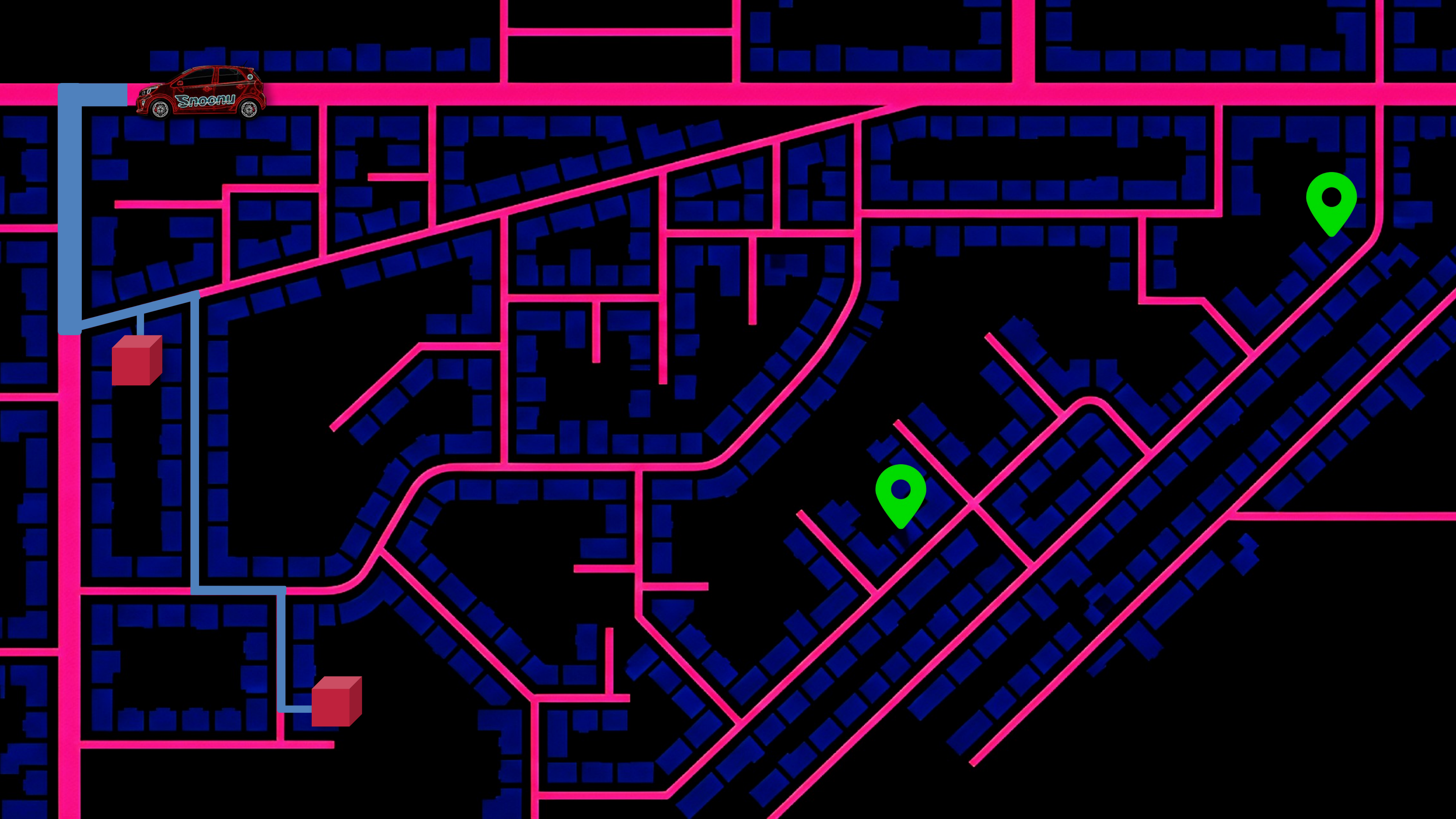

Umm Salal 'Alī
ام صلال على
Rawdat Al Faras
روضة الفرس
Al Nasraniya
النصرانية
Ash-Shahaniyah
الشحانية
Rawdat Al Hamama
Lusail
لوسيل
Umm Salal Muhammed
ام صلال
Umm Leghab
ام لقهاب
Umm Lekhya
ام لخيا
Umm Leghab
ام لقهاب
Lubara
لوبرة
Umm Al Afaei
أم الأفاعي
Lehsiniya
لحسنية
Rawdat Egdaim
روضة اقديم
Umm Al `Izam
أم اللزام
Wadi Al Majhoola
وادي المجهولة
Al-Rayyan
الريان
Doha
الدوحة
Umm Lebrak
ام لبرك
Rawdat Rashed
روضة راشد
Umm Al Zubar West
ام الزبار الغربية
Al Sahla Al Shamaliya
السهلة الشمالية
Umm Wishah
ام وشاح
Bu Yuwabi
Al Wukair
الوكير
Al Wakrah
الوكرة
Mekaines
مكينس
Abū Nakhlah
أبو نخله
Al Bahya
الباهية
Al Wakra South West
الوكرة جنوب غرب
Al Markhiya West
المرخية الغربية
Swaidat Al Warag
سويدات الورق
Umm Al Suwab
ام الصوب
Birkat Al Awamer
بركة العوامر
Mesaieed-Free Zone
مسيعيد - المنطقة الحرة

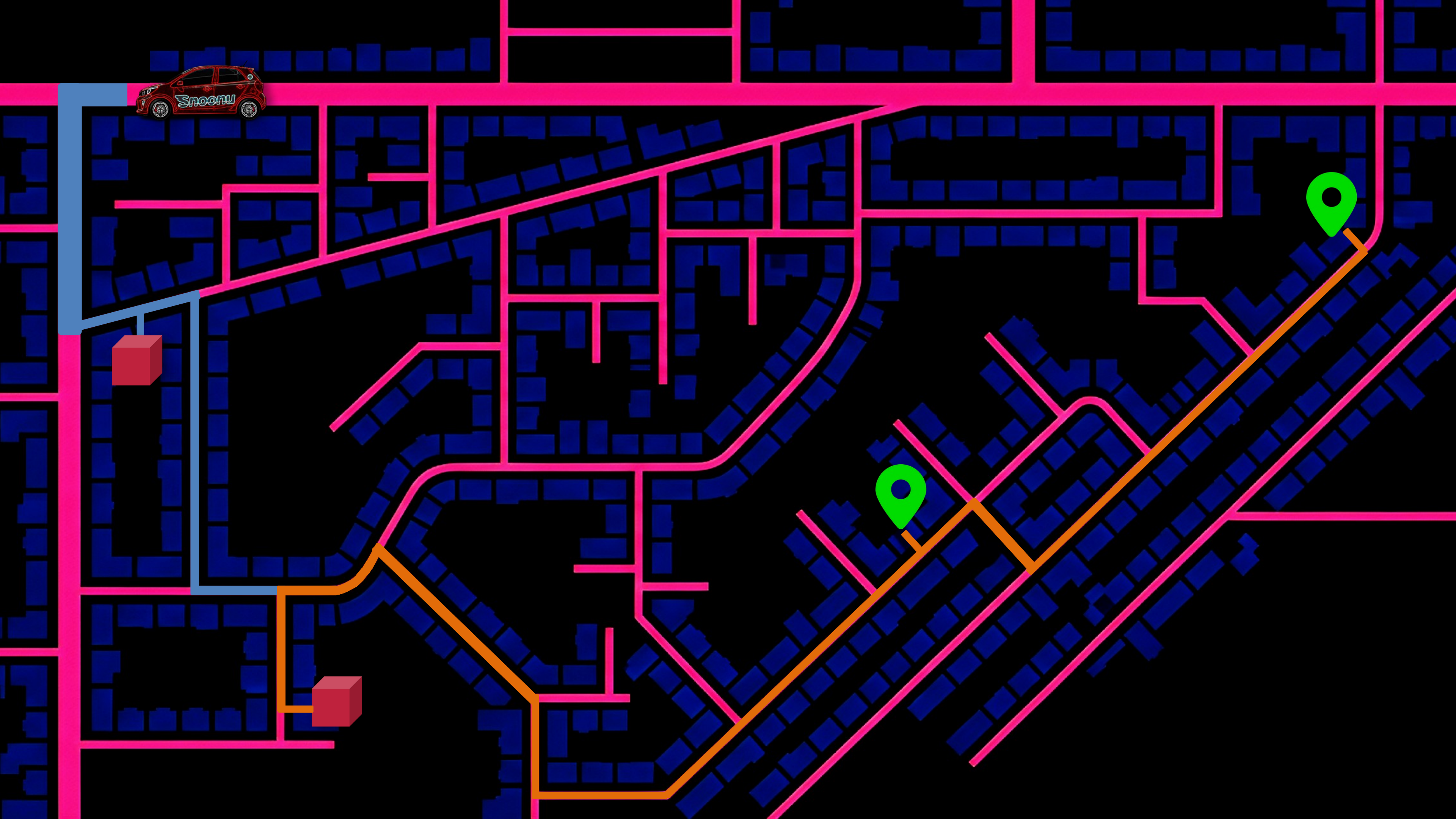


How can we REDUCE COSTS while maintaining delivery time and customer experience?

ORDERS BATCHING


Snoonu


Snoonu


Snoonu

Snoonu Hackathon

Your Path to Next-Gen Tech

واحة قطر للعلوم والتكنولوجيا
QATAR SCIENCE & TECHNOLOGY PARK
عضو في مؤسسة قطر
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QFALUMNI

مؤسسة قطر
Qatar Foundation

Smarter Batching, Scheduling and Dispatching for Faster Deliveries

Team: Sunaya, Phat, Tram, Jana

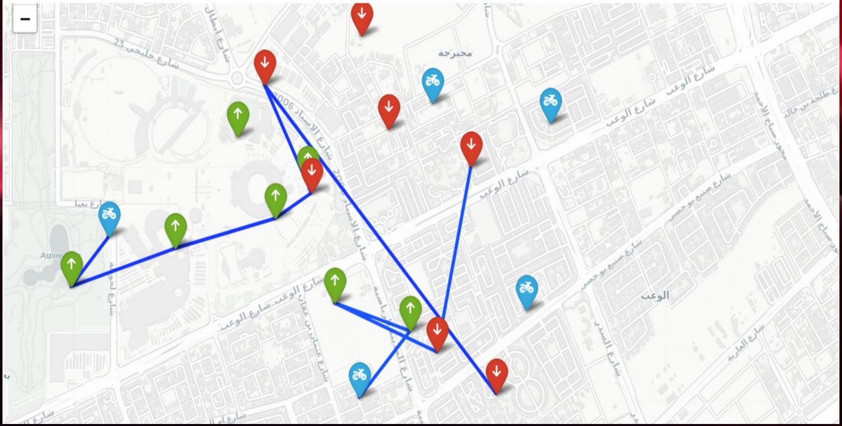
Evaluation Metrics

- Average Delivery Time
- Average Batch Size
- On-time Delivery Rate (ETA)
- Average Distance Travel (Petrol consumption)
- Driver Utilization (SD orders/driver)
- Time complexity of algorithm
- Customer Satisfaction

Sequential Single-Order Paths

Assign every order to the driver closest to it. For couriers assigned multiple orders, they complete each order sequentially (courier to pickup, then pickup to dropoff) before starting the next.

Provides a straightforward, unoptimized benchmark for comparison.



Greedy Nearest Neighbor

Overall reduction of

✓✗✎✚

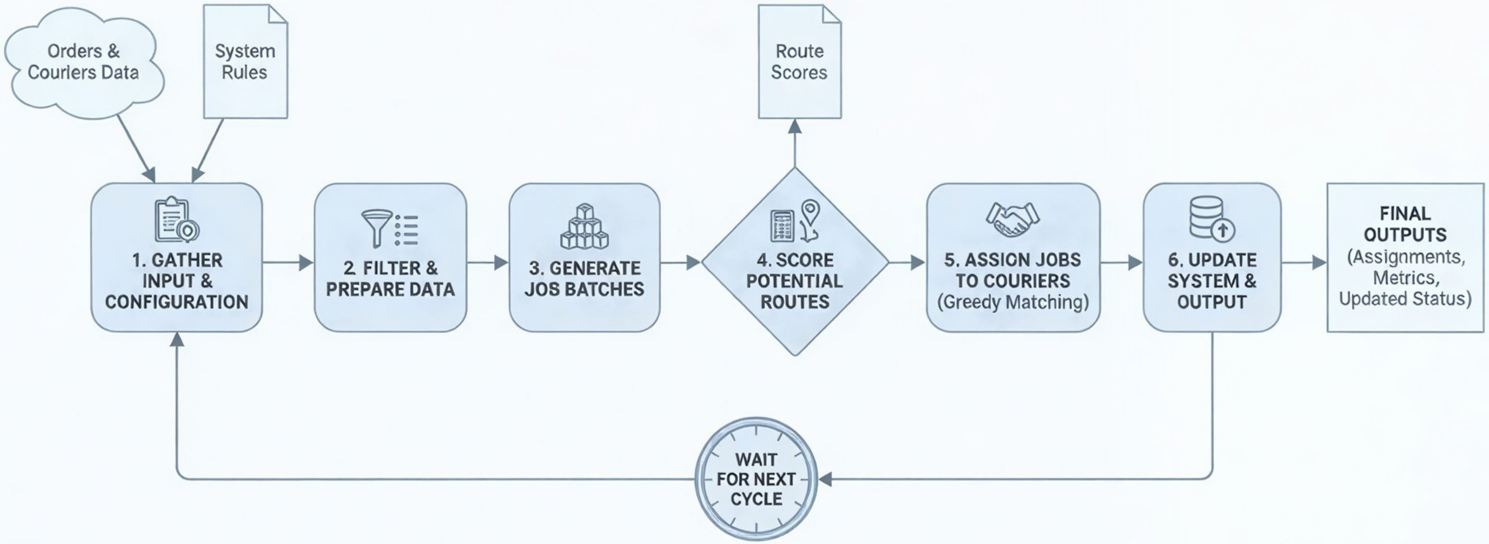
✗☎

in total travel distance

- This approach aims to optimize the delivery path for couriers handling multiple orders by dynamically selecting the next closest valid stop (either a pickup or a dropoff).
- Maintains a list of all pending pickups and dropoffs for the courier's assigned orders.
- Repeatedly selects the closest available stop.

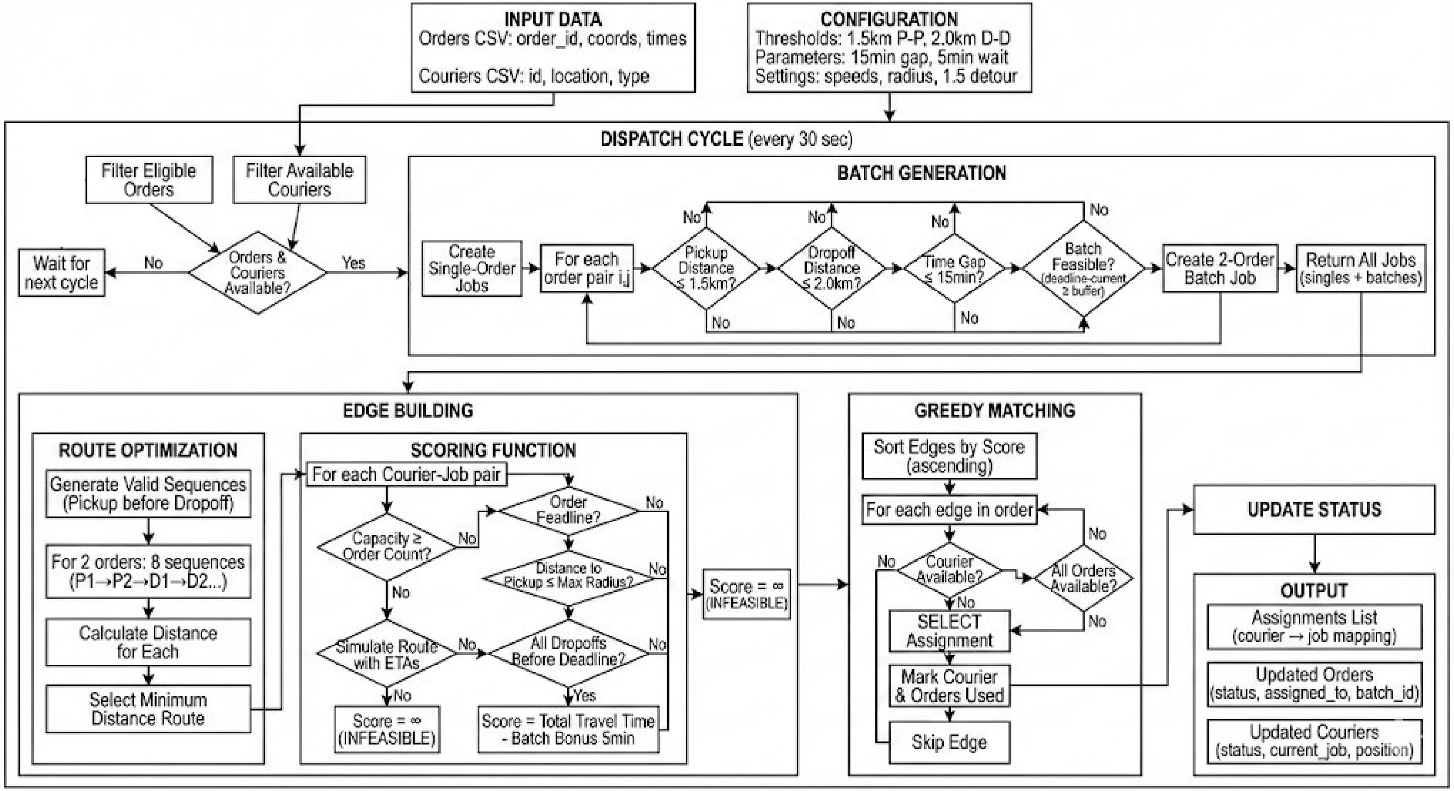
Algorithm

High-Level Dispatch Algorithm Overview



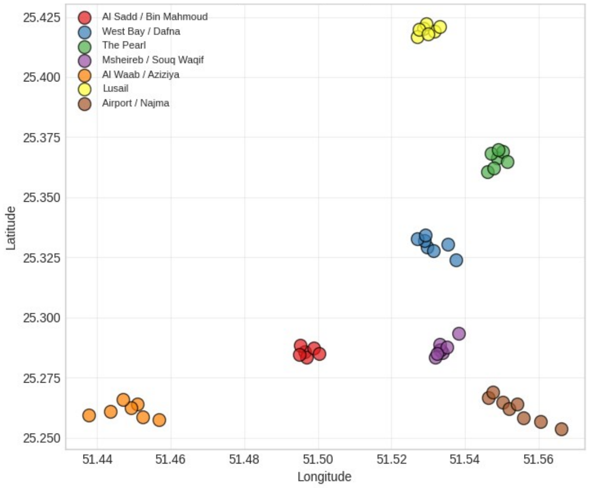
Time Complexity: $O(C * N^2 \log(C * N^2))$

where C is available courier and N is the number of orders

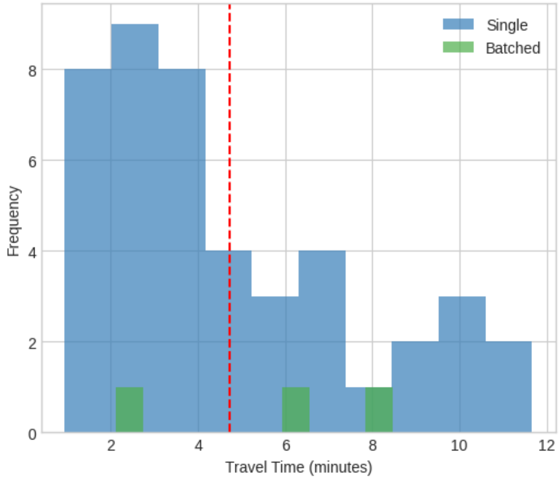

INPUT DATA
Orders CSV: order_id, coords, times
Couriers CSV: id, location, type
CONFIGURATION
Thresholds: 1.5km P-P, 2.0km D-D
Parameters: 15min gap, 5min wait
Settings: speeds, radius, 1.5 detour
DISPATCH CYCLE (every 30 sec)
Filter Eligible Orders
Filter Available Couriers
Wait for next cycle
No
Orders & Couriers Available?
Yes
BATCH GENERATION
Create Single-Order Jobs
For each order pair i,j
Pickup Distance ≤ 1.5km?
Dropoff Distance ≤ 2.0km?
Time Gap ≤ 15min?
Batch Feasible? (deadline-current 2 buffer)
Create 2-Order Batch Job
Return All Jobs (singles + batches)
No
EDGE BUILDING
ROUTE OPTIMIZATION
Generate Valid Sequences (Pickup before Dropoff)
For 2 orders: 8 sequences (P1→P2→D1→D2...)
Calculate Distance for Each
Select Minimum Distance Route
SCORING FUNCTION
For each Courier-Job pair
Capacity ≥ Order Count?
Order Feadline?
Distance to Pickup ≤ Max Radius?
Simulate Route with ETAs
All Dropoffs Before Deadline?
Yes
Score = ∞ (INFEASIBLE)
Score = Total Travel Time - Batch Bonus 5min
Score = ∞ (INFEASIBLE)
GREEDY MATCHING
Sort Edges by Score (ascending)
For each edge in order
Courier Available?
All Orders Available?
SELECT Assignment
Mark Courier & Orders Used
Skip Edge
UPDATE STATUS
OUTPUT
Assignments List (courier → job mapping)
Updated Orders (status, assigned_to, batch_id)
Updated Couriers (status, current_job, position)

Metrics

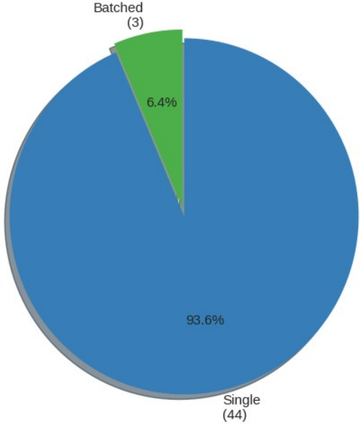
Pickup Locations by Geographic Cluster



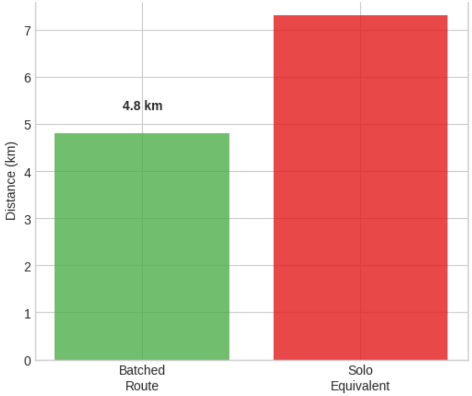
Travel Time Distribution



Assignment Types



Distance Savings 34.2% (7.3 km)



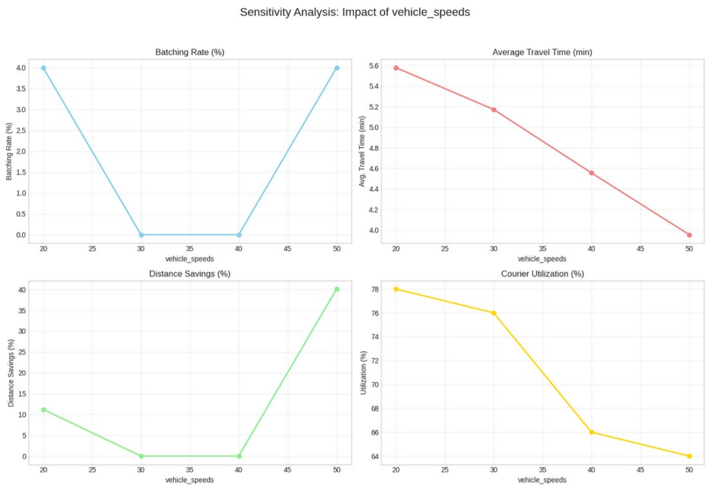
Ablation Tests

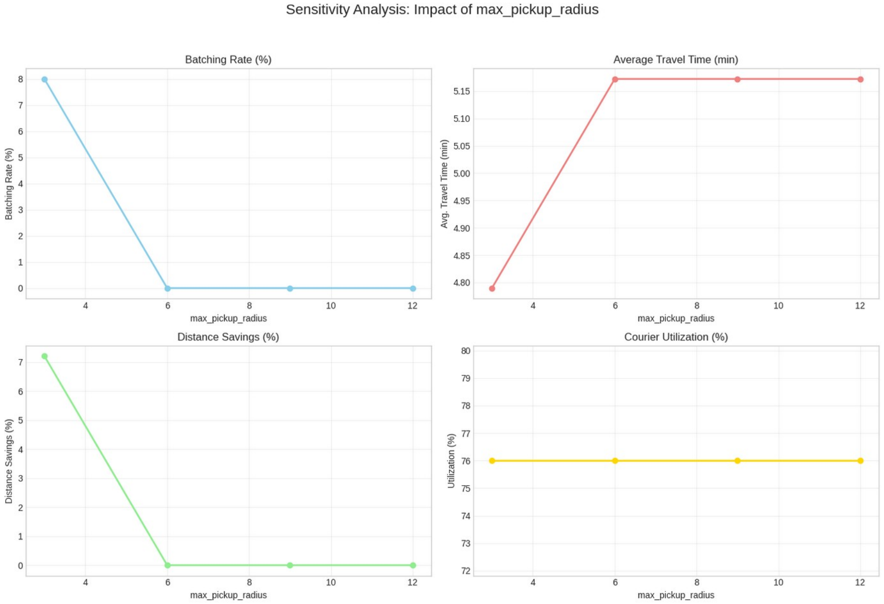
How do we choose the parameters?

Parameter	Optimal Value
`max_pickup_to_pickup_km`	10.0 km
`max_dropoff_to_dropoff_km`	2.0 km
`ready_time_threshold_min`	50 minutes
`max_detour_factor`	1.4
`max_batch_size`	3
`min_required_time_buffer`	3.0 minutes

Ablation Tests

How do we choose the parameters?





Future Work

Future Works

1. Predict where the next order will likely appear so idle couriers can reposition before orders arrive.
2. Use the expert planner that we developed to generate datasets to train a neural network (basically imitating the expert planner).
3. Reinforcement Learning (RL) optimizes online deliveries by training agents (systems) to make smart, dynamic decisions for couriers and platforms, learning from rewards (e.g., completed deliveries, reduced wait times) and penalties (e.g., missed orders, traffic).

Thank you for this opportunity.

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